

Marketing System of Some Commercially Important Marine Fish Species in the Chattogram Region of Bangladesh

3. MATERIALS AND METHODS

3.1 Description of the study area

The study area covers the marine fish marketing network of the Chattogram metropolitan region, the principal coastal trade hub of Bangladesh through which a large share of the country's marine landings passes before reaching inland markets (DoF, 2024). Chattogram is selected because it combines all structural segments of the marine fish channel within one urban economy: beach landing points, wholesale assembly markets, intermediary trading yards (arats), and dense retail neighbourhoods.

Six retail-and-wholesale fish markets (arats) constitute the empirical frame: Fishery Ghat and Patenga, both physically linked to a landing point so that first-sale prices can be observed directly; and Chawkbazar, Kazir Dewri, Karnaphuli Complex and Bahaddarhat, four retail markets distributed across old-town, central and newer residential areas so that consumer-side demand heterogeneity is represented. The final six were fixed using three criteria, direct landing-point access, wholesale-channel coverage, and geographic spread across the city and unlike a purely convenience-based selection, four of the six (Fishery Ghat, Chawkbazar, Kazir Dewri, Karnaphuli Complex) repeat the sites of the author's earlier 2019 baseline study, which supports a decade-level comparison, while Bahaddarhat and Patenga extend the frame to a newer residential retail market and a second landing point.

Table 3.1: Study markets, code prefixes and channel position

Market	Code	Channel position	Primary analytical contribution
Fishery Ghat	M1	Landing-linked wholesale-cum-retail	First-sale (producer-side) price observations
Chawkbazar	M2	Retail	Retail prices and trade volume quotes
Kazir Dewri	M3	Retail	Retail prices and vendor practices
Karnaphuli Complex	M4	Retail	Retail prices; comparability with the 2019 baseline
Bahaddarhat	M5	Retail	Newer-generation retail turnover
Patenga	M6	Landing-linked retail	Coast-side buy prices supporting producer-share estimation

3.2 Study period and research design

The investigation follows a descriptive-cum-analytical cross-sectional survey design executed within a ten-day operational window: one preparation day including pilot testing, approximately five consecutive days of primary data collection distributed over different weekday market sessions, one reserve day for weather or availability contingencies, and the

remaining days for data entry, verification and analysis. Field scheduling spreads each arat across different weekdays so that systematic differences between busy and slack trading sessions do not contaminate between-market price comparisons. The single-season snapshot character of this design is reported transparently as a limitation in Section 3.11, and temporal interpretation is restricted to the survey period.

The unit of inquiry is an individual market actor interviewed once, whereas the unit of price observation is one respondent-by-species quote expressed in Bangladeshi taka (BDT) per kilogram under fresh, ice-touched, whole-fish condition. Because respondents transact in heterogeneous lots, quantities quoted in maunds are converted using 1 maund = 37.32 kg, and monetary values are standardised to per-kilogram terms during data preparation. Prices refer to transactions on the interview day, supplemented by the previous high-volume day when trade was thin, and the reference day is recorded on every form.

3.3 Target population and sampling procedure

3.3.1 Population definition

The target population comprises all market intermediaries handling the selected marine fish species at the six study arats together with purchasing consumers at those arats. Three operator categories define the sampling strata. An **aratdar** operates the physical arat infrastructure, provides trading space plus services such as weighing facilities and auction coordination, and earns commission or lease income on transactions routed through his premises — he is a **commission-earning facilitator, not the owner of the fish**, which is carried through into the questionnaire design in Section 3.5. A **bepari/faria** is a mobile wholesaler who buys at landing points or upstream yards, transports consignments, and sells through arats either to retailers or to other wholesalers. A **khuchra (retail) vendor** sells directly to end-consumers from fixed stalls or vending spots inside the market.

3.3.2 Sample size determination

No complete census register of operators exists in these informal markets, so a planning benchmark is first obtained from Cochran's (1977) single-proportion formula, adopting a 95 percent confidence level ($Z = 1.96$), maximum response variance ($p = q = 0.5$), and an eight percent permissible error (relaxed from the conventional five percent given fixed field resources):

$$n_0 = Z^2 p q / e^2 = (1.96)^2(0.5)(0.5) / (0.08)^2 \approx 150 \dots\dots\dots (3.1)$$

As a robustness check, Yamane's (1967) finite-population formula, applied to an illustrative frame of $N = 1,000$ operators that plausibly exceeds the number of active intermediaries trading across six arats during any session, with the same $e = 0.08$:

$$n = N / (1 + N e^2) = 1,000 / (1 + 1,000 \times 0.0064) \approx 135 \dots\dots\dots (3.2)$$

Both benchmarks ($n_0 \approx 150$; $n \approx 135$) set an upper planning ceiling. It should be noted that Cochran's formula is designed for estimating a single population proportion; the study's actual inferential targets are continuous price and margin comparisons (Section 3.8), for

which a formal power calculation against an expected effect size would be the methodologically ideal approach. In the absence of a reliable prior effect-size estimate for this market, the proportion-based ceiling is used only as a planning bound, and the field target actually applied is fixed independently below, on resource and design grounds rather than on the ceiling itself.

Given the ten-day field window across six markets, the applied field target is 120 valid interviews, allocated equally across the three trader strata (5 aratdars + 5 beparis/farias + 5 khuchra retailers = 15 traders per market) plus 5 exit-intercepted consumers per market, across six markets (Table 3.3). This is below the Cochran ceiling of 150 but is deliberately chosen over an unequal, retail-heavy split so that each trader stratum carries equal analytical weight in the between-stratum comparisons of Section 3.8; the trade-off — fewer total interviews in exchange for balanced strata — is stated here explicitly rather than left implicit.

3.3.3 Allocation and selection of respondents

Table 3.3: Quota allocation of 120 respondents across the six markets

Market	Aratdar	Bepari/Faria	Khuchra retailer	Trader subtotal	Consumer	Total
M1 Fishery Ghat	5	5	5	15	5	20
M2 Chawkbazar	5	5	5	15	5	20
M3 Kazir Dewri	5	5	5	15	5	20
M4 Karnaphuli Complex	5	5	5	15	5	20
M5 Bahaddarhat	5	5	5	15	5	20
M6 Patenga	5	5	5	15	5	20
Total	30	30	30	90	30	120

Markets themselves constitute the primary stratum and were purposively nominated on the criteria in Section 3.1, because informal operator registers cannot provide an equal-probability frame. Within each selected market, aratdar and bepari/faria participants are drawn purposively among operators active during the survey session, prioritising informants handling the focal species, while khuchra vendors are taken successively along the market rows until the quota fills. Consumer observations use systematic exit interception of purchasers leaving the market with visible fish purchases. This stratified purposive-cum-quota procedure is disclosed honestly rather than presented as probability sampling, and inferential results are therefore interpreted as analytical generalisation within the studied system rather than statistical generalisation to a wider population.

3.3.4 Price matching protocol

To validate quoted margins against actually traded prices, each trader interview records the respondent's principal supply linkage — from whom the respondent purchased and to whom the respondent sold during the session. Whenever a retailer reports purchasing a surveyed species on the interview day and the corresponding seller is also interviewed, the transaction

receives a common pair identifier linking buyer and seller forms. Matched pairs enable Wilcoxon signed-rank validation of self-reported buy-versus-sell consistency and add an independent quality screen on stated margins. Given the compressed schedule, the number of genuinely matched pairs is expected to be modest (realistically a handful per market); this test should therefore be read as a diagnostic quality check rather than a fully powered hypothesis test, and the achieved number of matched pairs should be reported alongside the result.

3.4 Selection of fish species

Ten commercially important marine species form the analytical core of the study, selected against four criteria: persistent presence across all six survey arats during preliminary reconnaissance; substantial share in daily trade volume or value at Chattogram retail outlets; coverage of high-, medium- and low-price tiers so that margin behaviour can be compared across market segments; and documented importance in national marine fisheries statistics.

Scientific names throughout Table 3.4 follow FishBase (2026).

Table 3.4: 10 marine fish species selected for this study

Code	Local name	English common name	Scientific name	Tier
S01	Ilish	Hilsa shad	<i>Tenualosa ilisha</i>	High
S02	Rupchanda	Silver pomfret	<i>Pampus argenteus</i>	High
S03	Lakkha	Indian salmon (four-finger threadfin)	<i>Eleutheronema tetradactylum</i>	High
S04	Koral/Kurl	Asian sea bass	<i>Lates calcarifer</i>	High
S05	Surma	Indian mackerel	<i>Rastrelliger kanagurta</i>	Medium
S06	Churi	Largehead ribbonfish	<i>Trichiurus lepturus</i>	Medium
S07	Poa	Pama croaker	<i>Otolithoides pama</i>	Medium
S08	Kankoita	Thorny/Indian anchovy	<i>Stolephorus indicus</i>	Low-volume
S09	Loitta	Bombay duck	<i>Harpadon nehereus</i>	Low
S10	Harina	Chacunda gizzard shad	<i>Anodontostoma chacunda</i>	Low

Species S03 (previously Bagda chingri / giant tiger prawn, *Penaeus monodon*, in the initial candidate list) has been replaced with Lakkha (*Eleutheronema tetradactylum*), a high-value marine finfish, so that all ten entries are true fish and consistent with the study's title and scope; prawn is a crustacean and its inclusion in a 'marine fish species' study would be a taxonomic inconsistency. Species S08 is narrowed to *Stolephorus indicus* specifically, rather than a mixed anchovy genus group, so that each code maps to one identifiable species for per-species price analysis.

Should the pilot round reveal that any species is scarcely transacted at the study markets, a pre-specified substitution rule applies: the affected species is replaced by the closest-tier available alternative observed during the pilot, the change and its justification are recorded with date and supervisor acknowledgement before large-scale collection begins, and no

further substitutions occur thereafter so that the instrument remains stable throughout the survey.

3.5 Instruments and data collection procedure

Data were collected through pre-structured, interviewer-administered instruments prepared in Bangla to guarantee conceptual equivalence with respondents, accompanied by an English analytical specification for the researcher. The instrument battery comprises five coordinated forms whose items map one-to-one onto the variables defined in Section 3.6; nothing outside these fields was collected, in line with the minimised-data principle adopted for this study. Each physical Bangla form carries its Form ID (A, B, R, M or C) printed in its header so that field forms, data entry and this methodology stay unambiguously cross-referenced.

Trader questionnaires (Form A – Aratdar; Form B – Bepari/Faria; Form R – Khuchra retailer). A shared structural template underlies all three, with actor-specific filtering. Common modules capture basic demographics (age, education, duration in marine fish trade), principal supply linkage, quantities handled per day in kilograms, buy-and-sell quotations for each focal species, mode of payment (cash, bKash/Nagad, credit), and actor-specific cost modules. On Form A, the price-grid columns record the auction-clearing prices the aratdar facilitates — what the fisherman receives and what the bepari pays — as pass-through prices, not the aratdar's own trading margin; his income is captured separately as commission/lease income in the cost module, so that his margin is not double-counted once as a price spread and again as commission. The Form B module adds transport fares, icing quantity and expenditure, commission paid to aratdars, and transit damage percentages. Form R covers stall rent, icing cost, wash-water and tool expenses, self-reported spoilage percentages, and customer composition.

Market observation checklist (Form M). One checklist per market, completed by direct physical observation of twelve infrastructure characteristics (market type and hours, retail stall count, transport access, platform condition, roofing, drainage, electricity, ice availability, sanitation, water supply, and lease/toll arrangement), each marked with its evidence source (direct observation vs. verbal confirmation) plus GPS coordinates and photo references.

Consumer exit slip (Form C). Five purchasers leaving each market complete a two-to-three-minute intercept slip recording focal-species purchases with price paid, quantities, any other fish bought, and payment mode an independent consumer-side check on retailer sell prices.

Supplementary tag-price sheet (optional). Where field time permits, posted or spoken display prices of all fish observable at each market are transcribed once per visit, at zero interview burden, enriching the descriptive appendix without entering the formal hypothesis tests.

3.5.1 Interviewing schedule and pilot

Interviews are conducted during the morning trading window, when wholesale arrivals and early transactions concentrate, and extend into late-morning retail peaks at retail yards. Fieldwork opens with a one-day pilot at one market on approximately five respondents

covering every stratum: interviewer phrasing, skip-pattern integrity, species recognisability, maund-to-kilogram conversions and response times are checked, after which only supervisor-approved corrections are applied. Questionnaire and species-list versions are frozen before systematic collection begins.

3.6 Variables and their measurement

Table 3.5 summarises the measured variables, their originating instrument fields, measurement scales and intended analytical roles.

Table 3.5: Variable definitions, instrumentation and analytical roles

Variable	Instrument source	Scale / unit	Analytical role
Respondent identifiers	Header block, all forms	Codes M1–M6, actor class, serial	Stratification and linkage keys
Age; education; trade experience	Trader socio-economics module	Years; grade; years	Actor profile descriptive statistics
Quantity handled	Trader trade module	kg/day (maund converted)	Volume weights; cost normalisation
Buy price by species	Species grid (traders)	BDT/kg	Producer-side price proxy (P_p)
Sell price by species	Species grid (traders); Form C	BDT/kg	Retail price anchor (P_r); margin construction
Pair identifier	Link-tracing fields	Match code	Wilcoxon signed-rank validation sample
Marketing costs	Actor-specific cost modules	BDT + frequency codes	Per-kilogram marketing cost, MC
Commission	Form A, B modules	% per lot or BDT/kg	Channel margin decomposition
Damage / spoilage	Self-reported percentages	% of lot value	Loss adjustment; robustness note
Payment mode mix	Payment module, all forms	Cash / MFS / credit shares	Chi-square test of mode adoption
Market facilities	Form M observation	Graded checklist items	Contextual profile of arats
Tag prices	Optional sheet	BDT/kg, all species observed	Appendix descriptive diversity table

3.7 Analytical framework for marketing performance

3.7.1 Marketing cost

For every trader i in stratum k , total marketing cost per kilogram is the sum of itemised cash expenses (rent/toll, electricity where applicable, transport including loading/unloading, wages, commission paid, icing, tools and washing water, and other sundries), normalised to the kilogram basis by quantity handled:

$$MC_{ki} = (\sum c_{ri}) / Q_{ki}, r = 1 \dots R \dots\dots\dots (3.3)$$

where c_{ri} = expense on cost item r by respondent i ; Q_{ki} = quantity handled per day by respondent i in stratum k ; R = number of applicable cost items.

3.7.2 Segment margins and profit

The absolute margin earned at any channel segment equals the spread between the respondent's recorded selling and buying prices for the same species-day quote, and segment profit deducts marketing cost from that spread:

$$M_{ki} = P_{si} - P_{bi} \dots\dots\dots (3.4)$$

$$\pi_{ki} = M_{ki} - MC_{ki} \dots\dots\dots (3.5)$$

where P_{si} and P_{bi} = selling and buying prices per kilogram quoted by respondent i ; π_{ki} = net profit of the segment operator per kilogram.

3.7.3 Channel-level indicators

Following the marketing-margin framework of Kohls and Uhl (1980), four standard chain-level indicators are computed per species and market combination, using mean prices over valid observations, where P_r = mean retail price per kilogram (validated by consumer slips) and P_p = mean producer-side, first-sale price per kilogram:

$$\text{Price spread} = P_r - P_p \dots\dots\dots (3.6)$$

$$GMM\% = [(P_r - P_p) / P_r] \times 100 \dots\dots\dots (3.7)$$

$$NMM\% = [(P_r - P_p - \sum MC) / P_r] \times 100 \dots\dots\dots (3.8)$$

$$PS\% = (P_p / P_r) \times 100 \dots\dots\dots (3.9)$$

where $\sum MC$ = summed per-kilogram marketing costs of all surveyed segments in the chain; $GMM\%$ = gross marketing margin percentage; $NMM\%$ = net marketing margin percentage; $PS\%$ = producer's share of the consumer's taka.

3.7.4 Marketing efficiency

Operational efficiency follows Shepherd's (1965) ratio of output value to input costs, applied on a per-kilogram chain basis; values above unity indicate that more of the final price remains as product value than is consumed by marketing services:

$$ME = P_r / (\sum MC + \sum M) \dots\dots\dots (3.10)$$

where ME = Shepherd-type efficiency index; $\sum M$ = summed segment margins along the chain. As a complementary producer-referenced measure, following Acharya and Agarwal's (1987) convention that efficiency assessment should also be judged from the producer's side, the same ratio is recomputed with P_p (producer price) in place of P_r ; the two versions are reported side by side rather than treated as interchangeable, since they answer different

questions (value delivered to the consumer per unit cost, versus value retained by the producer per unit cost).

3.8 Statistical analysis

Analysis runs in R statistical software (R Core Team, 2025) with an annotated script retained for full reproducibility. Price and margin distributions in informal fish markets rarely satisfy normality or homogeneity of variance confirmed here by Shapiro-Wilk screening on the pooled series before any test is applied, so the inferential battery relies on distribution-free procedures throughout, evaluated two-sidedly at the five percent significance level. Table 3.6 maps each research question onto its variable set and designated test, fixing the analytic plan before data inspection.

Table 3.6: Analytical questions mapped to statistical tests and decision rules ($\alpha = 0.05$)

Research question	Data	Test statistic	Decision rule
Do prices of the same species differ across markets?	Species-wise price observations by market ($n \geq 5$ cells)	Kruskal-Wallis H (Kruskal & Wallis, 1952); post-hoc Dunn (Dunn, 1964) with Holm adjustment	Reject H_0 when $p < 0.05$
Do margins differ across trader categories?	Segment margins by stratum	Mann-Whitney U per pair of strata	Reject H_0 when $p < 0.05$
Are self-reported buy-sell quotes internally consistent?	Matched identifiers	pair Wilcoxon signed-rank on paired differences	Non-significance corroborates consistency
Does payment-mode adoption differ across actor types?	Mode mixes by class	actor Chi-square; Fisher fallback for sparse cells	Reject H_0 when $p < 0.05$
Is marketing cost associated with segment profit?	Per-kilogram versus π pairs	MC Spearman rank correlation	Sign and strength of rho, 95% CI
Normality screening preceding all tests	Pooled price and margin series	Shapiro-Wilk W	$p > 0.05$ needed to justify parametric alternatives; else nonparametric used

Multiple pairwise contrasts following significant omnibus tests use Holm's (1979) step-down familywise error control. Because species compositions differ across respondents, inter-market comparisons are always executed within a single species, never pooled across species. Species appearing at fewer than three markets with fewer than five observations per cell are reported descriptively only.

3.9 Data quality assurance

Interviewers receive one structured training session covering consent administration, species coding, maund-to-kilogram conversion, skip patterns and the pair-linking rule, immediately

followed by the pilot round (Section 3.5.1). Completed forms are checked same-day against arithmetic range constraints, and daily field debriefs resolve anomalies while memory remains fresh. Approximately ten percent of trader respondents are re-approached through short callback verification of key fields (quantity handled, commission rates). During entry, respondent identifiers are screened for duplication, pair identifiers are reconciled across linked forms, and implausible entries are flagged rather than deleted so that exclusion decisions remain documented and traceable. Buy-price records that traders decline to disclose retain their analytical role under an explicitly flagged proxy convention approved before analysis.

3.10 Ethical considerations

Participation is voluntary and based on verbal informed consent:

- I. the interviewer reads a standard statement in Bangla explaining the study purpose, the academic-only use of information, anonymity protections, and the right to decline any question or withdraw entirely without consequence;
- II. personal identifiers are replaced by coded IDs at collection time, contact details are not recorded on forms, and electronic files reside exclusively on a password-protected device accessible to the researcher;
- III. reported results aggregate to group level so that no individual operator can be identified from published tables or figures;
- IV. the protocol follows the academic norms of the departmental term-paper course, and supervisor endorsement of instruments constitutes the institutional review of the survey plan.

3.11 Limitations of the methodology

Five limitations bound the interpretation of findings.

- a) The cross-sectional design captures one seasonal episode; margins in marine markets vary intra-week and seasonally, so temporal extrapolation beyond the survey window is avoided, and any seasonal claim in the discussion should be sourced from prior published data rather than presented as a within-study finding.
- b) Purposive nomination of markets combined with quota-convenience selection within strata precludes design-based population inference; results generalise analytically to comparable channel settings rather than statistically to all Chattogram operators.
- c) Producer-side prices are proxied from first-sale quotations at landings-linked markets rather than from boat-level fisher interviews, and this proxy status is carried explicitly into the producer-share calculations.
- d) Cost and loss variables rest partly on respondent recall and self-assessment, especially spoilage percentages; day-referencing of quantities and percentage-format questioning mitigate but cannot eliminate recall error.

- e) The field sample size (120) was fixed on resource and stratum-balance grounds rather than derived from a power calculation matched to the study's actual continuous-variable tests; the Cochran/Yamane figures in Section 3.3.2 should be read as a proportion-based planning ceiling, not as a formal power guarantee for the Kruskal-Wallis/Mann-Whitney comparisons in Section 3.8.

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